# The Data Design Processes

*DATA 3302: Data Visualization*

***Professor Austin P. Wright***
Cal Poly San Luis Obispo · Fall 2026

Slides based on Munzner, Visualization Analysis and Design, 2014

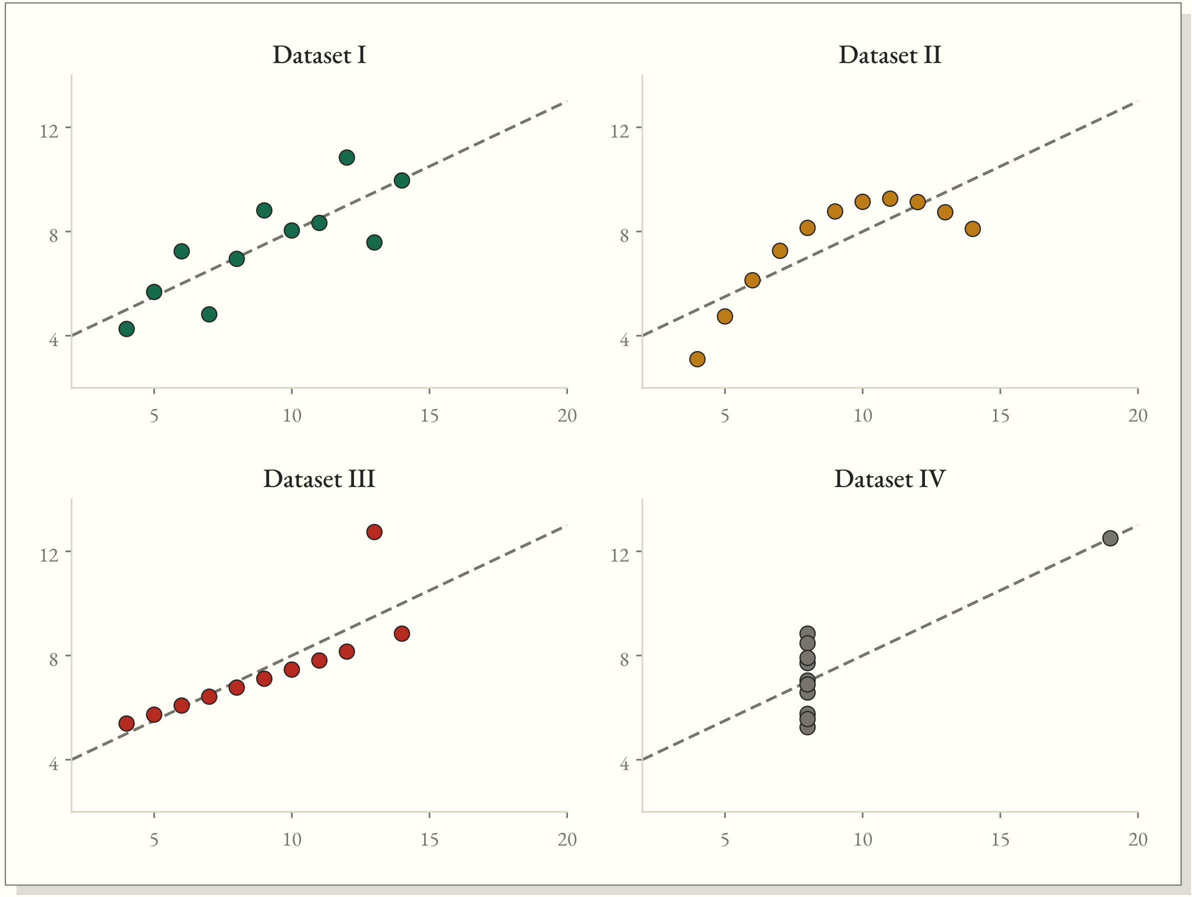
# *Why involve humans?*

# *Why involve computers?*

# *Why involve representations?*

# *Anscombe's Quartet*

| | *I* | *II* | *III* | *IV* |
|---|---:|---:|---:|---:|
| Mean of $x$ | 9.00 | 9.00 | 9.00 | 9.00 |
| Mean of $y$ | 7.50 | 7.50 | 7.50 | 7.50 |
| Variance of $x$ | 11.00 | 11.00 | 11.00 | 11.00 |
| Variance of $y$ | 4.13 | 4.13 | 4.12 | 4.12 |
| Correlation | 0.82 | 0.82 | 0.82 | 0.82 |
| Regression Fit | $y = 3 + 0.5x$ | $y = 3 + 0.x$ | $y = 3 + 0.5x$ | $y = 3 + 0.5x$ |

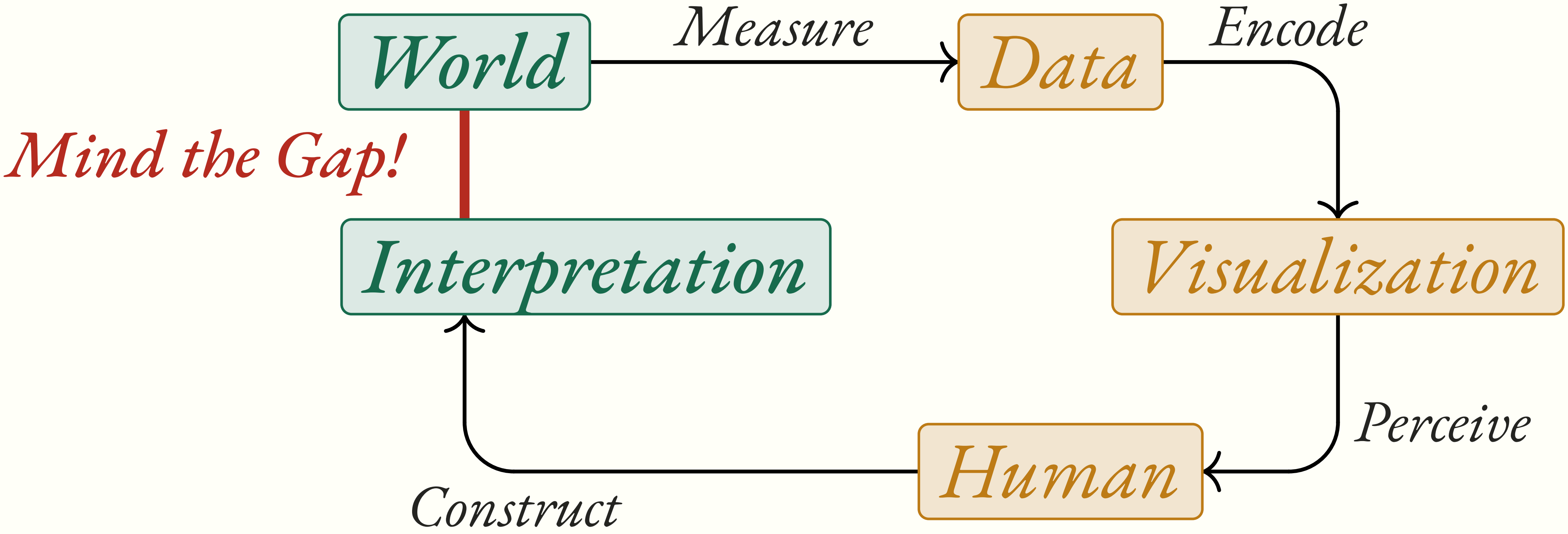



Anscombe, “Graphs in Statistical Analysis,” 1973

# *Why depend on vision?*

*Why not on summarize?*

## *Why is this difficult?*


World
Measure
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Mind the Gap!
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Visualization
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Human
Construct

# *Data Abstraction*

# *Conceptual Model vs Data Model*

## *Data Types*

1. Items / Hard Index
2. Attributes / Variables
3. Links / Relations
4. Positions / Soft Index
5. Grids / Topologies

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## *Dataset Types*

1. Tables
2. Networks
3. Fields
4. Geometry
5. Sets

### *Dataset Availability*

1. Static / Offline
2. Dynamic / Online

### *Dataset Mutabilty*

1. Fixed / Immutable
2. Reactive / Mutable

# A City Bike-Share System

Riders check out *bikes* from *stations* scattered across downtown. Every *trip* links a start station to an end station. Each station reports its *GPS coordinates*.

Your task is to build a continuously updated citywide *demand heatmap*. You start by identifying the data abstractions.

> *Identify the Types*
>
> 1. Dataset Type?
> 2. Data Types?
> 3. Dataset Availability/Mutability?

# *Attribute Types*

## *Attribute Types*

1. Categorical / Nominal
2. Ordinal
3. Quantitative (Interval)
4. Quantitative (Ratio)

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1. Categorical / Nominal
2. Ordinal
3. Quantitative (Interval)
4. Quantitative (Ratio)

## *Order Directions*

1. Sequential
2. Diverging
3. Cyclic

| A | B | C | S | T | U |
|---|---|---|---|---|---|
| Order ID | Order Date | Order Priority | Product Container | Product Base Margin | Ship Date |
| 3 | 10/14/06 | 5-Low | Large Box | 0.8 | 10/21/06 |
| 6 | 2/21/08 | 4-Not Specified | Small Pack | 0.55 | 2/22/08 |
| 32 | 7/16/07 | 2-High | Small Pack | 0.79 | 7/17/07 |
| 32 | 7/16/07 | 2-High | Jumbo Box | 0.72 | 7/17/07 |
| 32 | 7/16/07 | 2-High | Medium Box | 0.6 | 7/18/07 |
| 32 | 7/16/07 | 2-High | Medium Box | 0.65 | 7/18/07 |
| 35 | 10/23/07 | 4-Not Specified | Wrap Bag | 0.52 | 10/24/07 |
| 35 | 10/23/07 | 4-Not Specified | Small Box | 0.58 | 10/25/07 |
| 36 | 11/3/07 | 1-Urgent | Small Box | 0.55 | 11/3/07 |
| 65 | 3/18/07 | 1-Urgent | Small Pack | 0.49 | 3/19/07 |
| 66 | 1/20/05 | 5-Low | Wrap Bag | 0.56 | 1/20/05 |

*Example Dataset of Orders with Attribute Types*

# *Data Semantics*

## *Abstract Semantics*

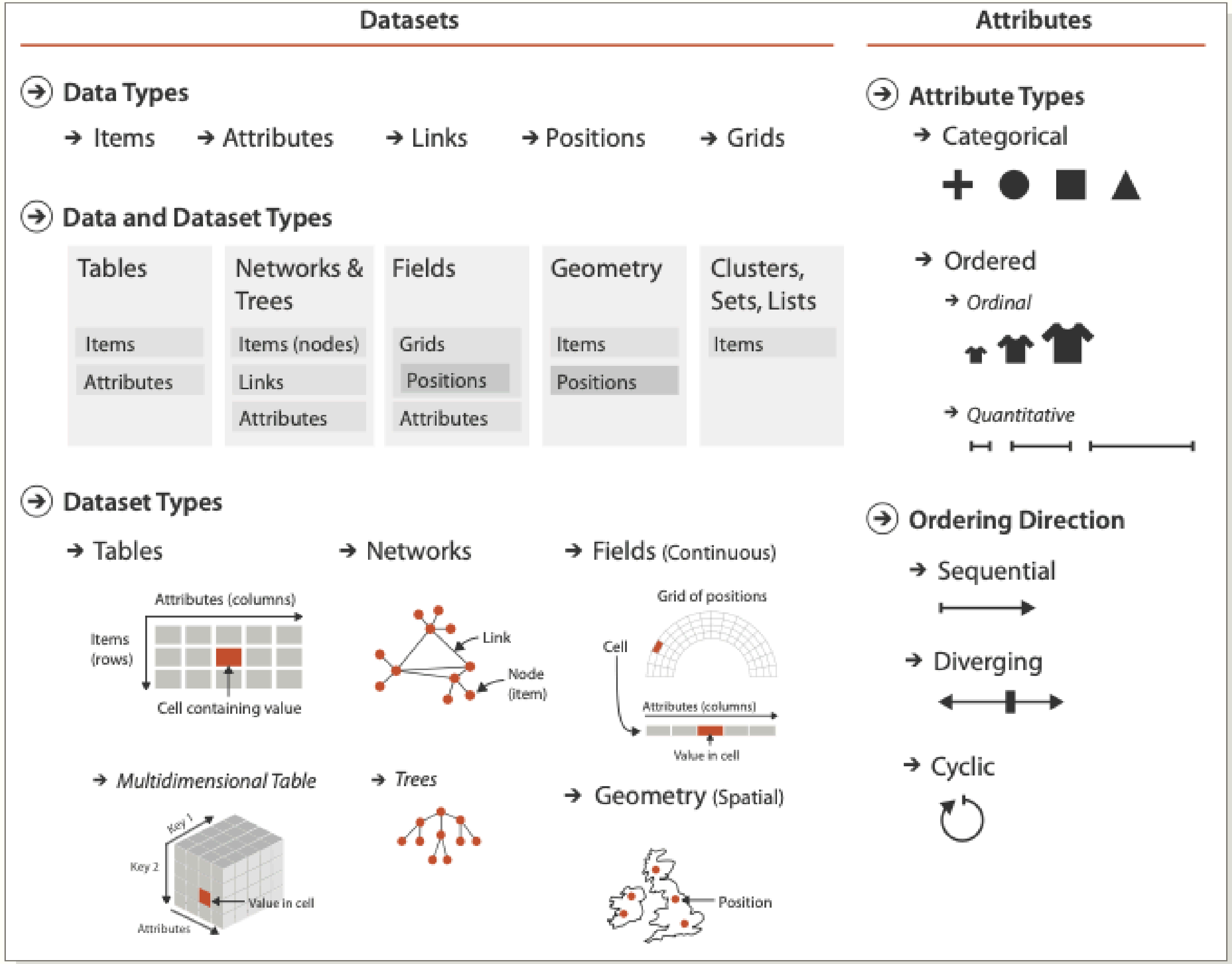
1. Key / Index
2. Value / Measure

### *Abstract Semantics*

1. Key / Index
2. Value / Measure

### *Structural Semantics*

1. Spatial
2. Temporal
3. Field
4. Vector




Munzner 2014, Figure 2.1

# *Data Transformations and Tidy Data*

*Data is rarely ready to visualize right away, you'll often need to* ***transform***, ***merge***, *or* ***tidy*** *it first.*

# The Relational Model

## *Relational Data*

1. *Table*: Dataset
2. *Row*: Record
3. *Column*: variable (name + type)
4. *Schema*: Table's column names & types
5. *Database*: Set of tables

| *Team* | *Wins* | *Losses* |
|---|---|---|
| Warriors | 10 | 4 |
| Kings | 7 | 7 |
| Lakers | 3 | 11 |

Each ***row*** is a record, each ***column*** is a variable, each ***cell*** is a value.

# Relational Algebra

Table(s) in, table out.

### *Core Operations*

1. *select* — choose a set of columns
2. *filter* — remove unwanted rows
3. *order by* — sort records by column values
4. *group by* + aggregate (`sum`, `mean`, ...) — partition, then summarize
5. *join* / *union* — combine multiple tables

Implemented by SQL, pandas, polars, *arquero*, and similar table libraries.

# Select & Filter

| *A* | *B* | *C* |
|---|---|---|
| x | 1 | 4 |
| y | 2 | 5 |
| z | 3 | 6 |

```
select “A”, “B”
```

| *A* | *B* |
|---|---|
| x | 1 |
| y | 2 |
| z | 3 |

```
filter “B” % 2 = 1
```

| *A* | *B* | *C* |
|---|---|---|
| x | 1 | 4 |
| z | 3 | 6 |

# Group By + Aggregate

```
group by "A"; aggregate mean
```

Original → Aggregated

| *A* | *B* | *C* |
|---|---|---|
| x | 1 | 4 |
| y | 2 | 5 |
| z | 3 | 6 |
| x | 7 | 9 |
| x | 8 | 10 |
| y | 11 | 12 |

| *A* | *B* | *C* |
|---|---|---|
| x | 5.33 | 7.67 |
| y | 6.5 | 8.5 |
| z | 3 | 6 |

# Combining Tables: Join

`inner join` $T_1$`,` $T_2$ `on “A”`

$T_1$

| $A$ | $B$ | $C$ |
|---|---|---|
| x | 1 | 4 |
| y | 2 | 5 |
| z | 3 | 6 |

$T_2$

| $A$ | $D$ | $E$ |
|---|---|---|
| x | 7 | 10 |
| why | 8 | 11 |
| z | 9 | 12 |

$\longrightarrow$

Result

| $A$ | $B$ | $C$ | $D$ | $E$ |
|---|---|---|---|---|
| x | 1 | 4 | 7 | 10 |
| z | 3 | 6 | 9 | 12 |

An ***inner join*** keeps only matching keys. An ***outer join*** keeps every key filling in missing values.

Based on this, what would a ***left join*** keep?

# *Tidy Data*

# Tidy Data

> ***Rules of Tidy Data***
>
> 1. Each *observation* is a row
> 2. Each *variable* is a column
> 3. Each *value* is a cell

| *month* | *temp* | *precip* | *wind* |
|---|---|---|---|
| January | 56.55 | 0.06 | 5.32 |
| February | 56.10 | 0.12 | 5.46 |
| March | 56.36 | 0.11 | 6.52 |

Tidy data relies on ***both*** a conceptual model (what's counts as an observation?) and a data model.

# Wide vs Narrow Tables

*Better for data entry vs data analysis*

*Tidy*

| *month* | *temp* | *precip* | *wind* |
|---|---|---|---|
| January | 56.55 | 0.06 | 5.32 |
| February | 56.10 | 0.12 | 5.46 |

*Untidy*

| *month* | *element* | *value* |
|---|---|---|
| January | temp | 56.55 |
| January | precip | 0.06 |
| January | wind | 5.32 |
| February | temp | 56.10 |
| February | precip | 0.12 |
| February | wind | 5.46 |

In the untidy table, a single observation is spread across several rows, and the ***value*** column mixes different variables. Reshaping between these forms is called ***pivoting*** and ***unpivoting***.

# Discussion

Is this table tidy? What are its variables and its observations?

| *artist* | *track* | *date_entered* | *wk1* | *wk2* | *wk3* |
|---|---|---|---|---|---|
| 2 Pac | Baby Don't Cry | 2000-02-26 | 87 | 82 | 72 |
| 3 Doors Down | Kryptonite | 2000-04-08 | 81 | 70 | 68 |
| 98° | Give Me Just One Night | 2000-08-19 | 51 | 39 | 34 |

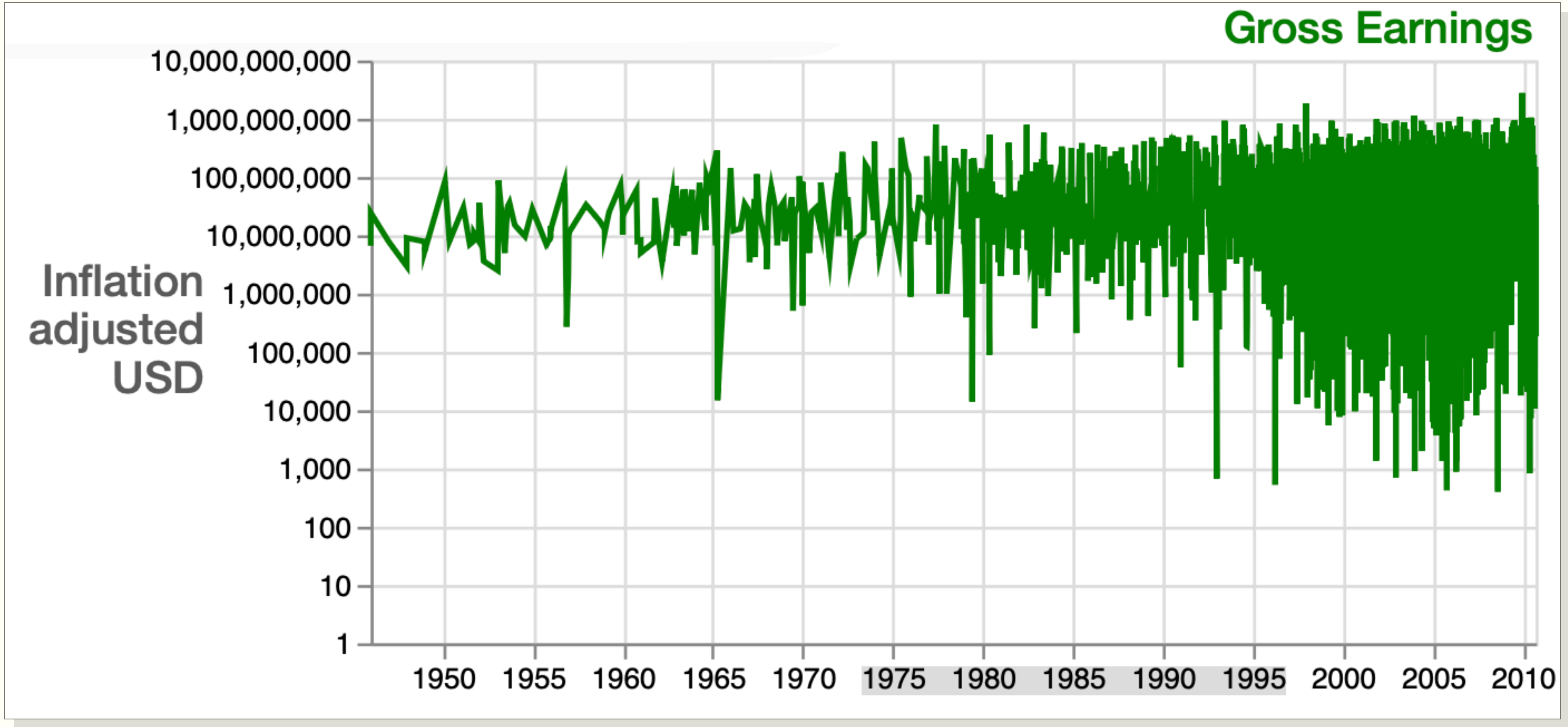
Weekly Billboard ranks, 2000 · *R for Data Science*, Hadley Wickham

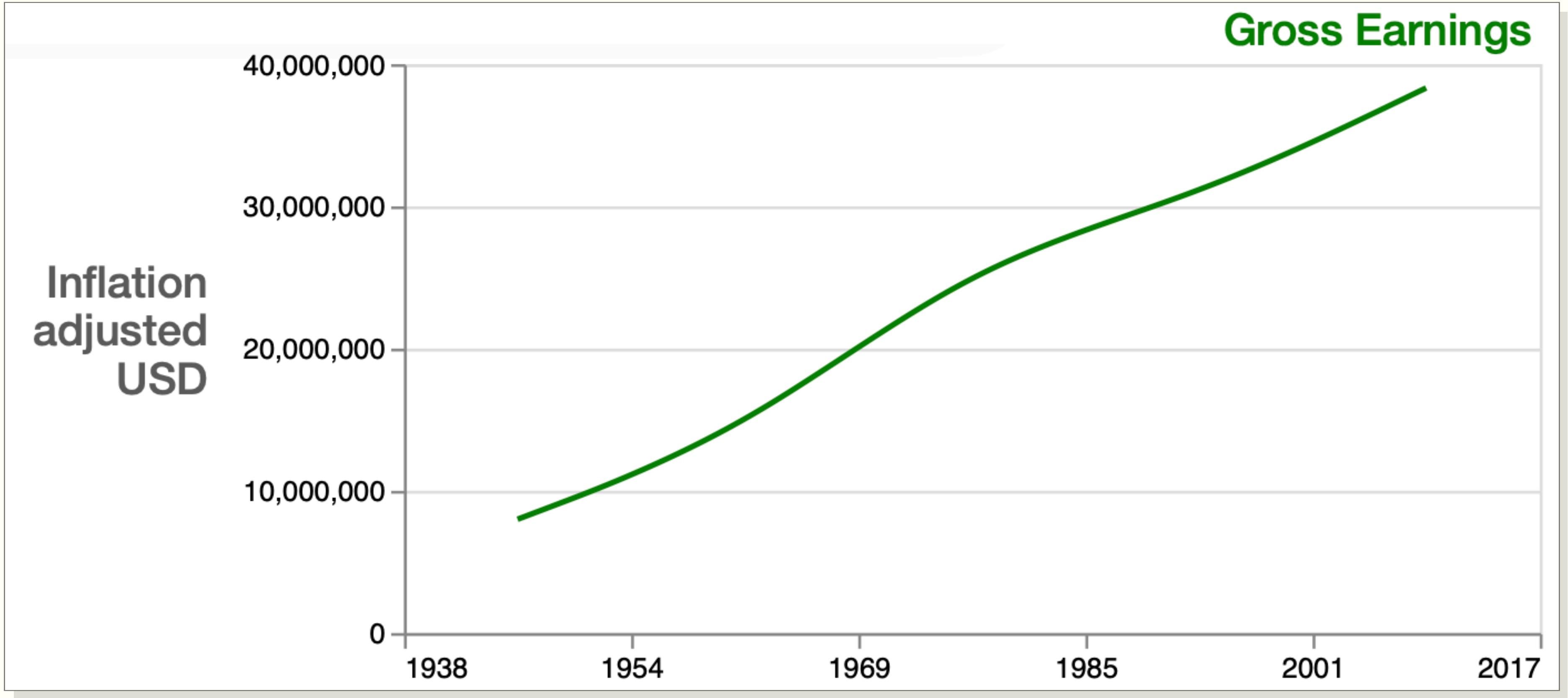
*One person's tidy is another person's mess*

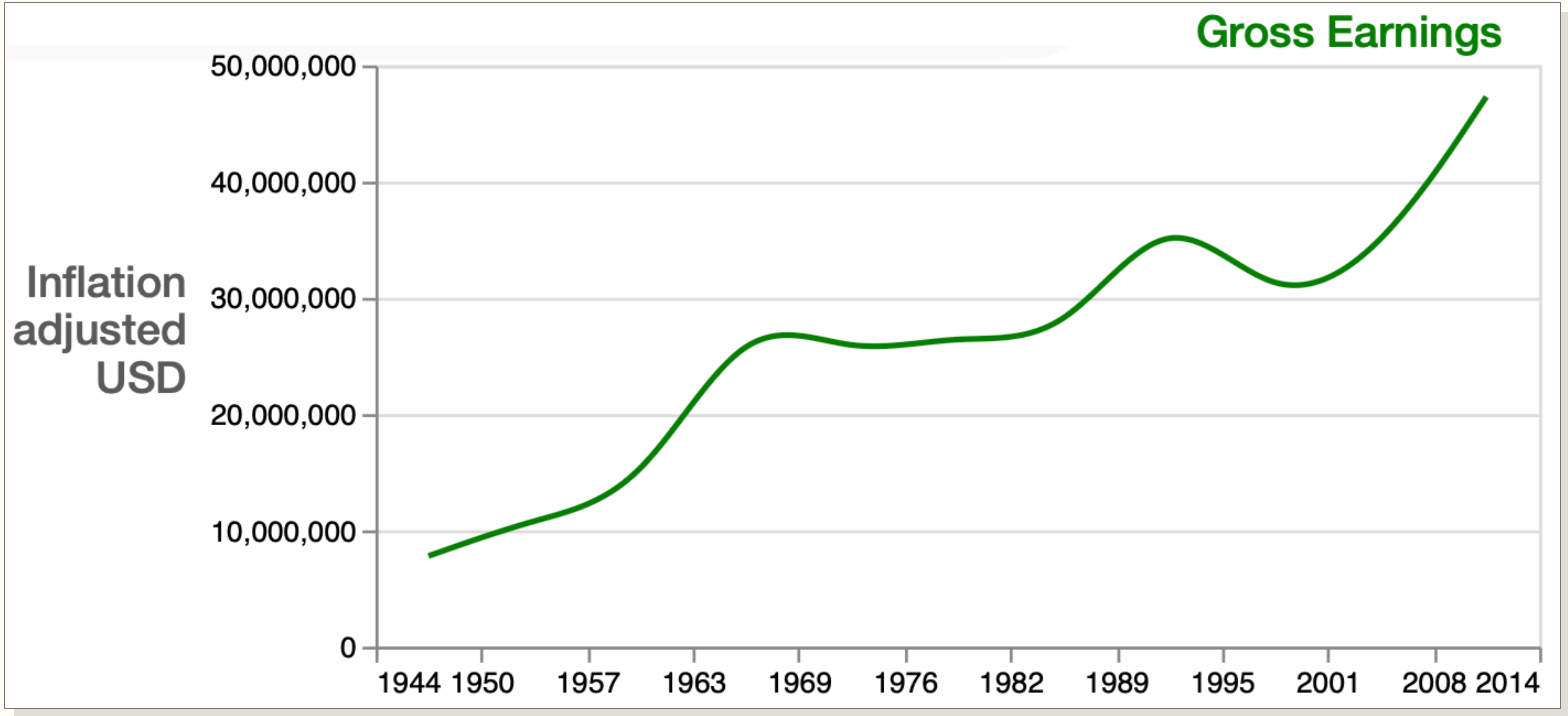
> *This means that most real analyses will require at least a little tidying. You'll begin by figuring out what the underlying **variables** and **observations** are. Sometimes this is easy; other times you'll need to consult with the people who originally generated the data.*
>
> — *Hadley Wickham*
>
> *R for Data Science*, "Data Tidying"

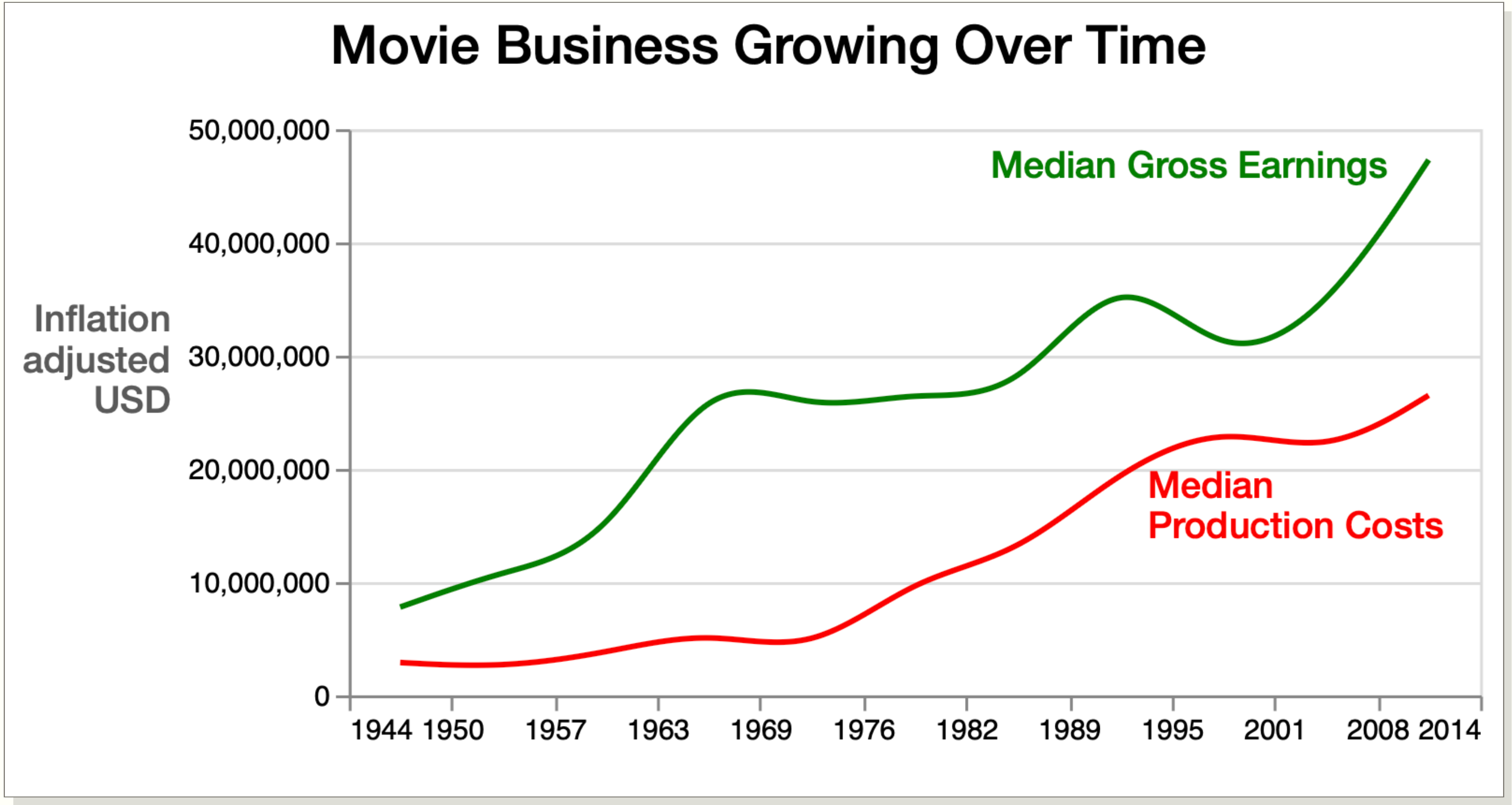
# *Visualization (In)Dependent Transformations*

| Title | Release_Date | Worldwide_Gross | Production_Budget | IMDB_Rating |
|---|---|---|---|---|
| The Land Girls | Jun 12 1998 | 146083.0 | 8000000.0 | 6.1 |
| First Love, Last Rites | Aug 07 1998 | 10876.0 | 300000.0 | 6.9 |
| I Married a Strange Person | Aug 28 1998 | 203134.0 | 250000.0 | 6.8 |
| Let's Talk About Sex | Sep 11 1998 | 373615.0 | 300000.0 | NaN |
| Slam | Oct 09 1998 | 1087521.0 | 1000000.0 | 3.4 |
| ... | ... | ... | ... | ... |
| Zack and Miri Make a Porno | Oct 31 2008 | 36851125.0 | 24000000.0 | 7.0 |
| Zodiac | Mar 02 2007 | 83080084.0 | 85000000.0 | NaN |
| Zoom | Aug 11 2006 | 12506188.0 | 35000000.0 | 3.4 |
| The Legend of Zorro | Oct 28 2005 | 141475336.0 | 80000000.0 | 5.7 |
| The Mask of Zorro | Jul 17 1998 | 233700000.0 | 65000000.0 | 6.7 |

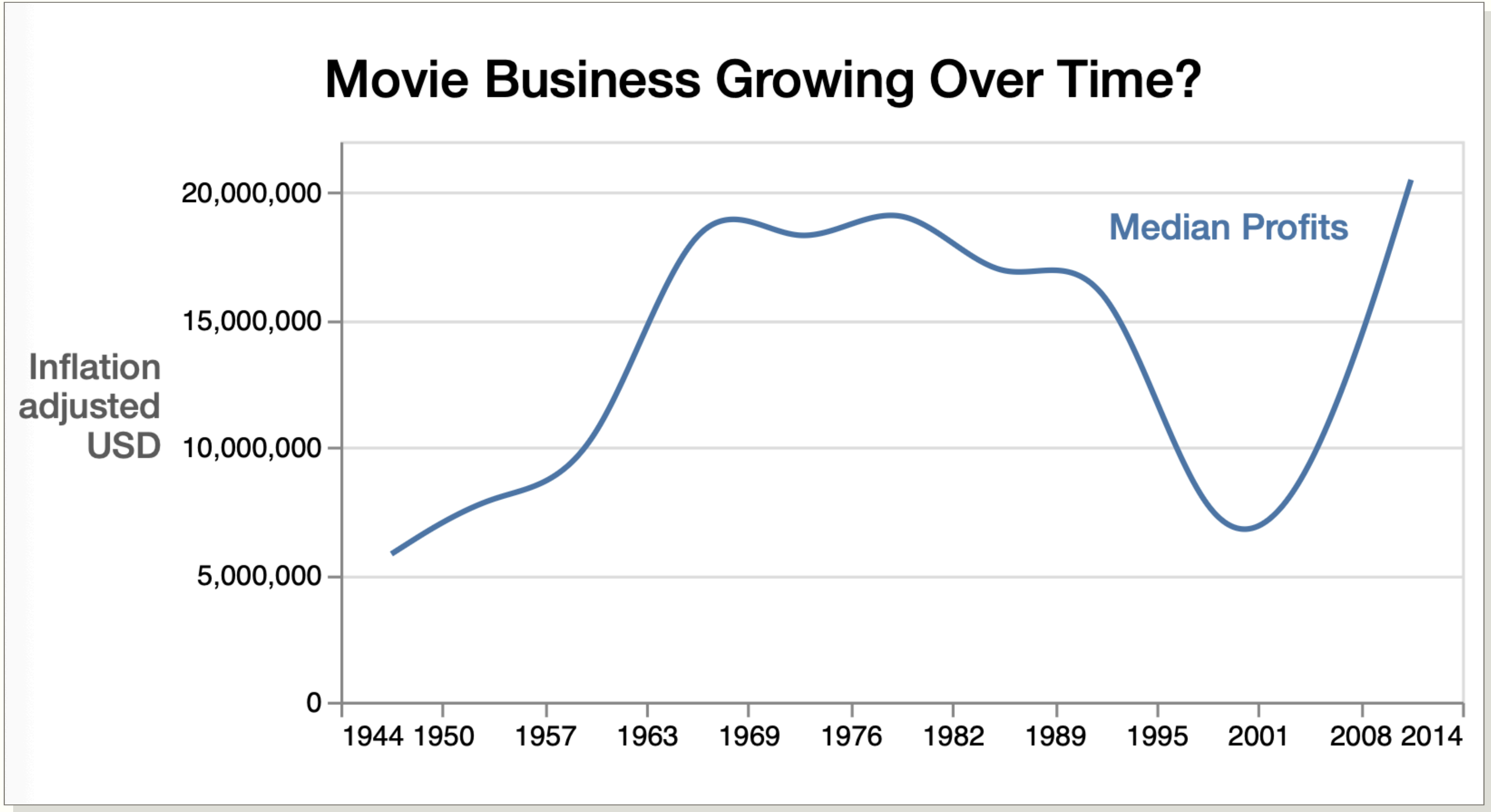

Gross Earnings
Inflation adjusted USD
10,000,000,000
1,000,000,000
100,000,000
10,000,000
1,000,000
100,000
10,000
1,000
100
10
1
1950
1955
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010


Gross Earnings
Inflation adjusted USD
40,000,000
30,000,000
20,000,000
10,000,000
0
1938
1954
1969
1985
2001
2017


Gross Earnings
Inflation adjusted USD
50,000,000
40,000,000
30,000,000
20,000,000
10,000,000
0
1944 1950 1957 1963 1969 1976 1982 1989 1995 2001 2008 2014



Movie Business Growing Over Time
Median Gross Earnings
Median
Production Costs
Inflation
adjusted
USD
50,000,000
40,000,000
30,000,000
20,000,000
10,000,000
0
1944 1950 1957 1963 1969 1976 1982 1989 1995 2001 2008 2014


Movie Business Growing Over Time?
Median Profits
Inflation adjusted USD
20,000,000
15,000,000
10,000,000
5,000,000
0
1944 1950 1957 1963 1969 1976 1982 1989 1995 2001 2008 2014

# *Task Abstraction*

*What does the user actually do with the visualization?*

# *User Actions*

## *Analyze*

1. Consume
    1. Discover
    2. Present
    3. Enjoy

# *User Actions*

## *Analyze*

1. Consume
    1. Discover
    2. Present
    3. Enjoy
2. Produce
    1. Annotate
    2. Record
    3. Derive

# User Actions

## Analyze

1. Consume
    1. Discover
    2. Present
    3. Enjoy
2. Produce
    1. Annotate
    2. Record
    3. Derive

## Search

1. Target Known
    1. Lookup
    2. Locate
2. Target Unknown
    1. Browse
    2. Explore

# *User Actions*

## *Analyze*

1. Consume
    1. Discover
    2. Present
    3. Enjoy
2. Produce
    1. Annotate
    2. Record
    3. Derive

## *Search*

1. Target Known
    1. Lookup
    2. Locate
2. Target Unknown
    1. Browse
    2. Explore

## *Query*

1. Identify
2. Compare
3. Summarize

# *User Actions (verbs) vs User Targets (nouns)*

# *User Targets*

## *Full Dataset*

1. Trends
2. Features
3. Outliers / Anomalies

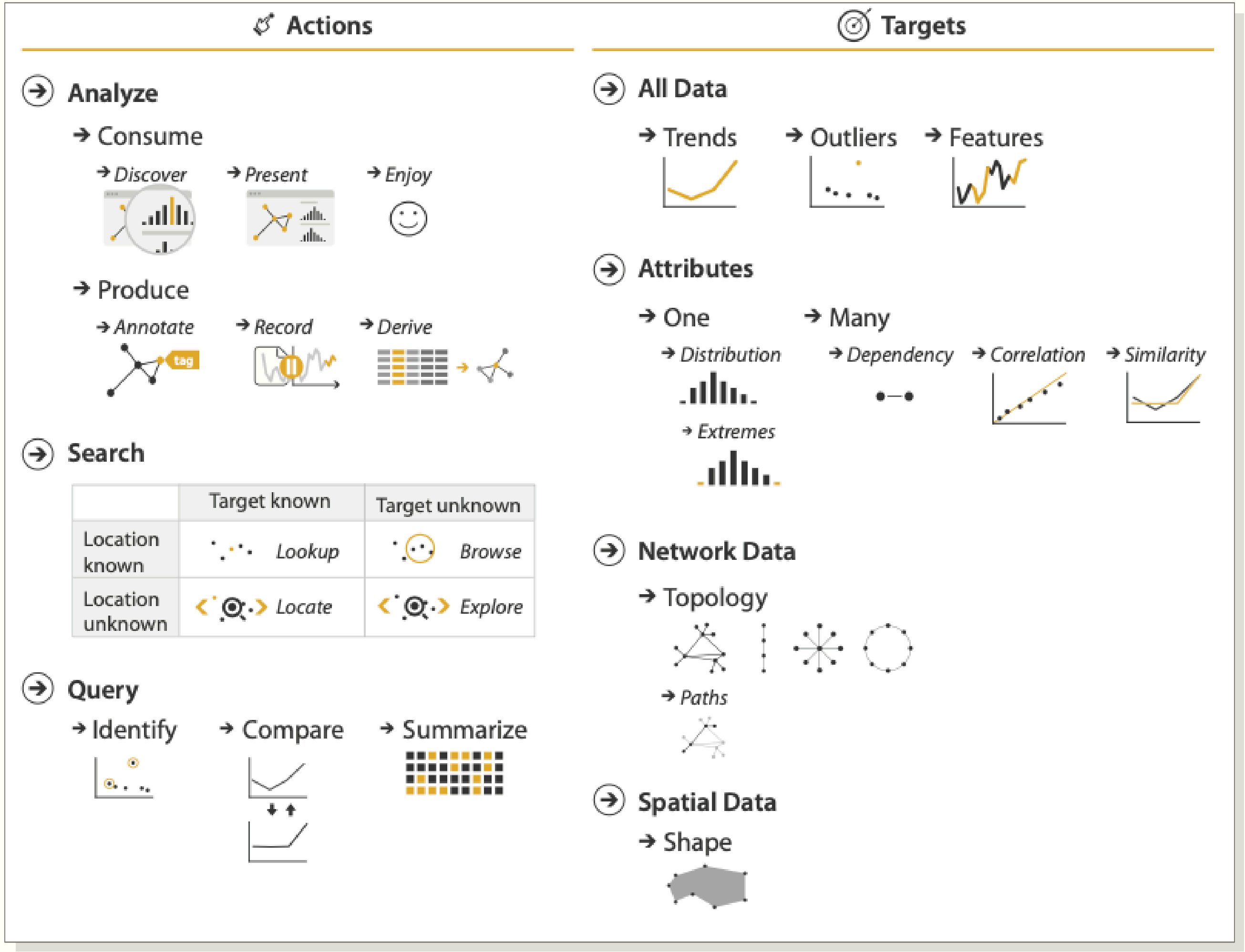
# *User Targets*

## *Full Dataset*

1. Trends
2. Features
3. Outliers / Anomalies

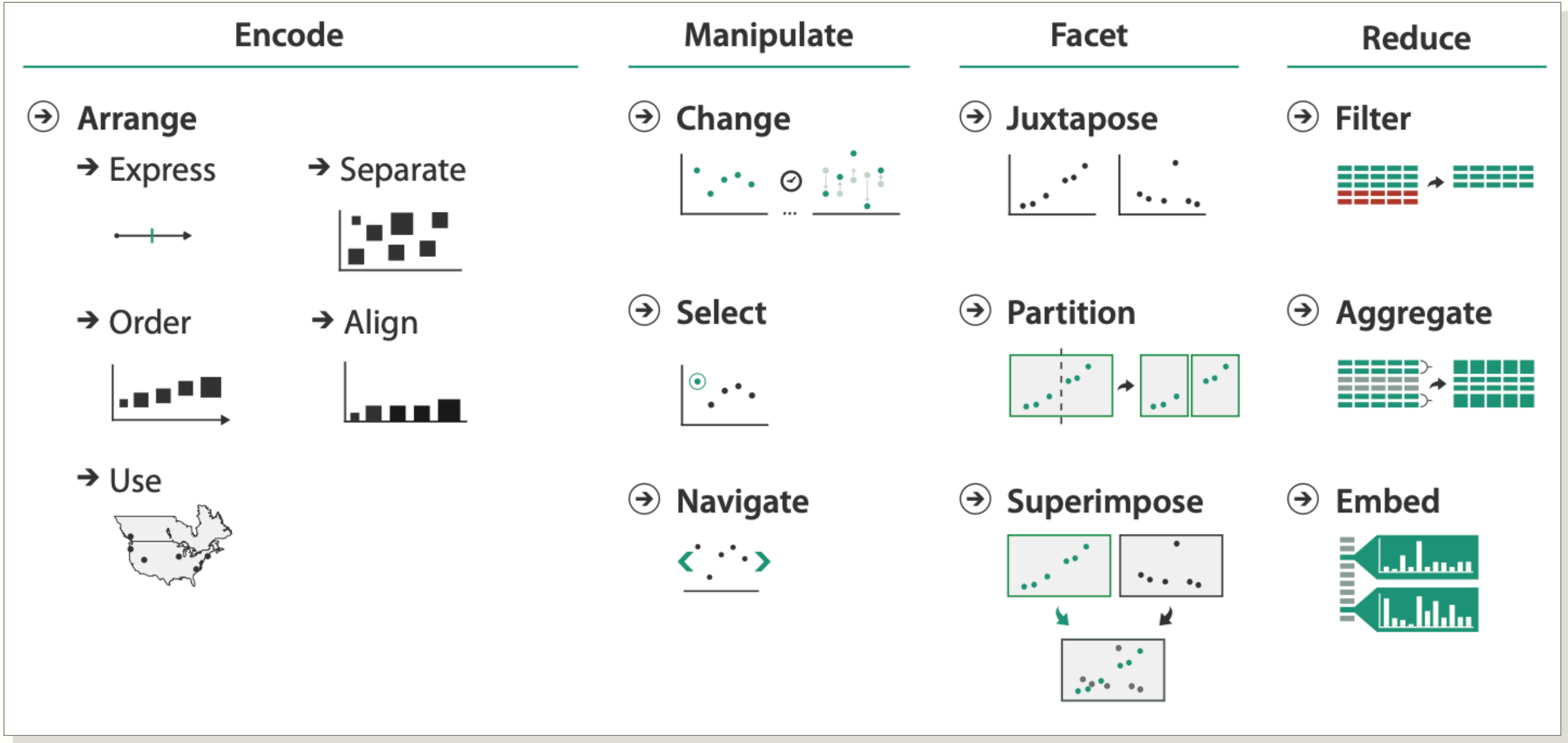
## *Attributes*

1. Single Attribute
    1. Distribution
    2. Paradigms
    3. Extremes
2. Multiple Attributes
    1. Dependency
    2. Correlation
    3. Similarity

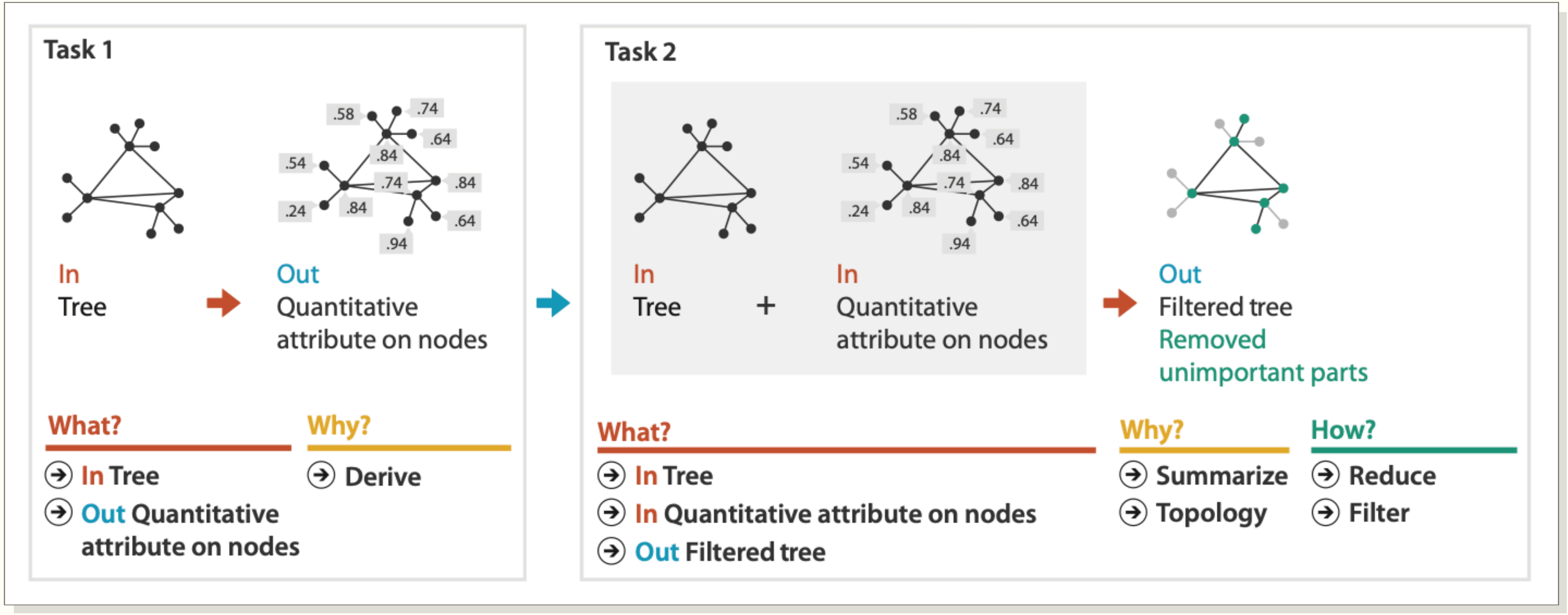



Munzner 2014, Figure 3.1

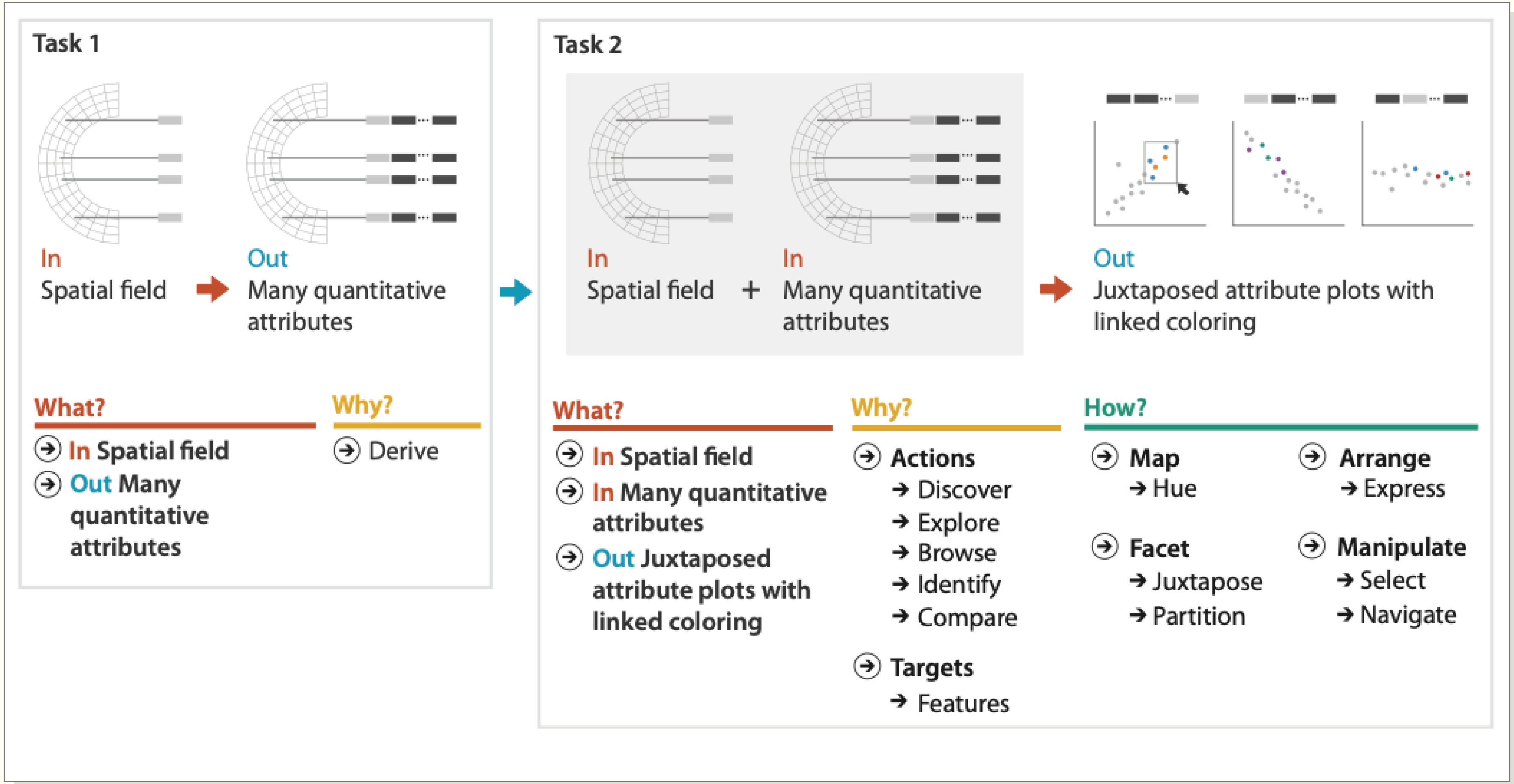
*How do designs support user tasks?*



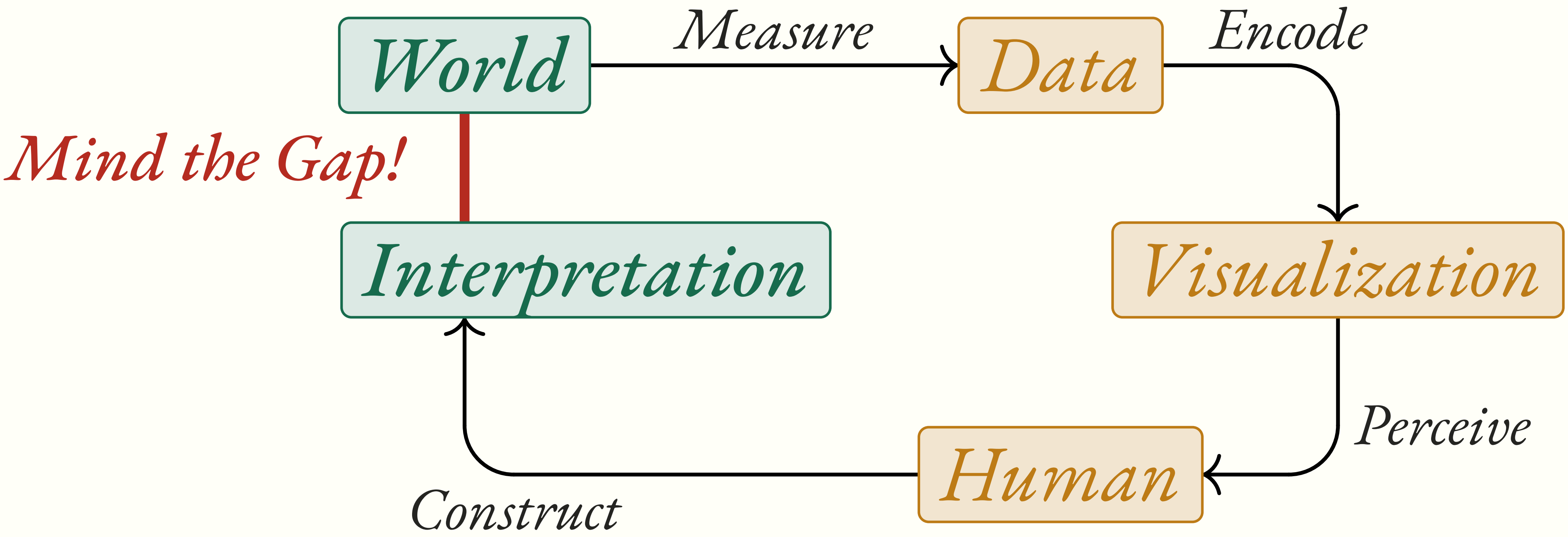

Munzner 2014, Figure 3.7




Munzner 2014, Figure 3.11




Munzner 2014, Figure 3.13


World
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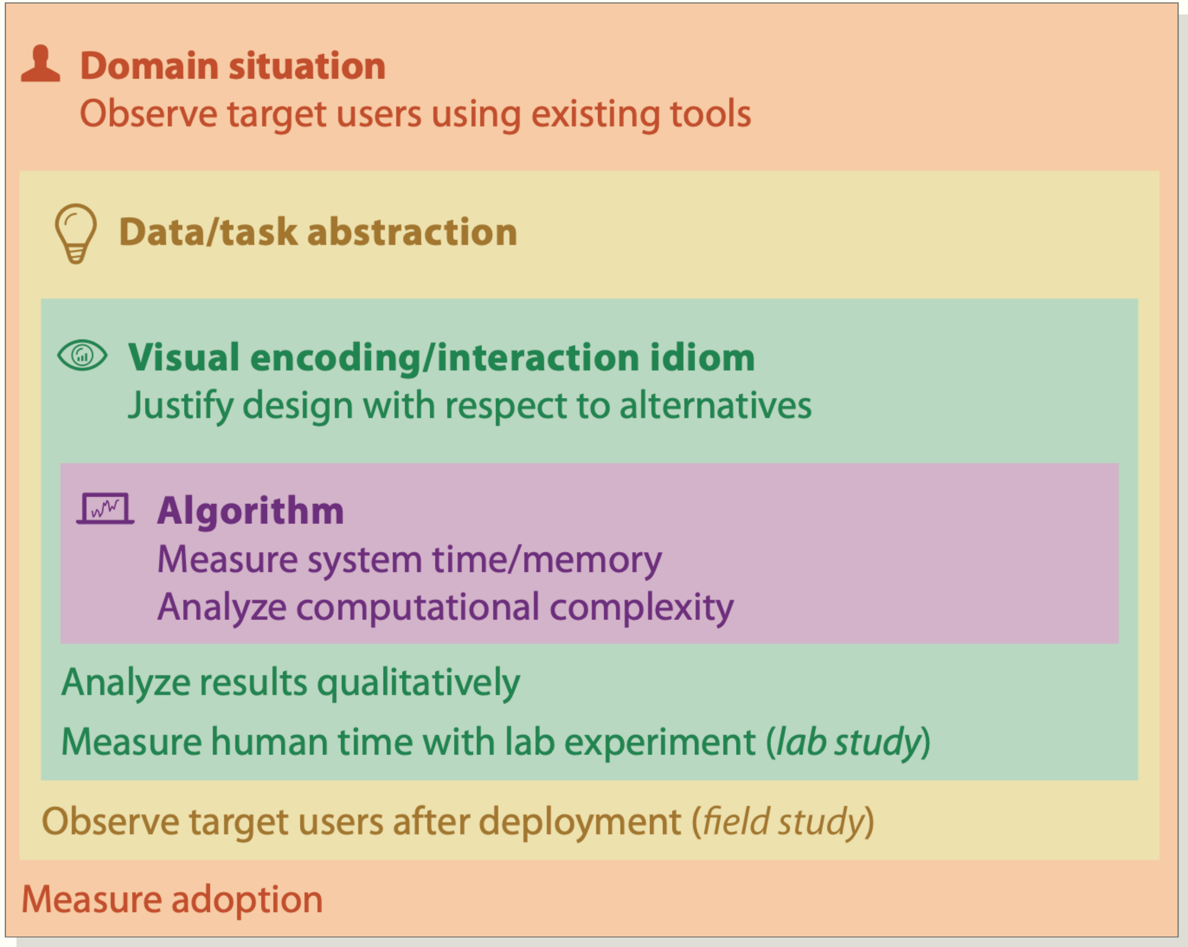
*How do we know if we did a good job?*

# *The Nested Model*




Munzner 2014, Figure 4.1

*How do we actually make good designs?*

# *The Design Process*

*Based on slides from Tal Wolman*

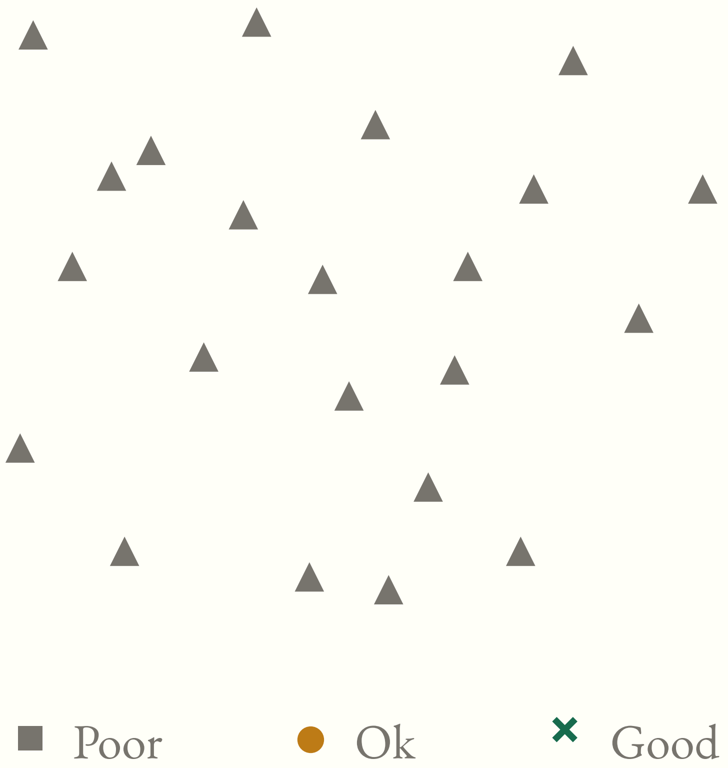
> *The design space of possible visualizations is huge, and includes considerations of both* ***how to create*** *and* ***how to interact*** *with visual representations. Vis design is full of trade-offs, and most possibilities in the design space* ***are ineffective*** *for a particular task — designers must weigh the resource limitations of computers, of humans, and of displays.*

— *Tamara Munzner*

Visualization Analysis & Design, 2014

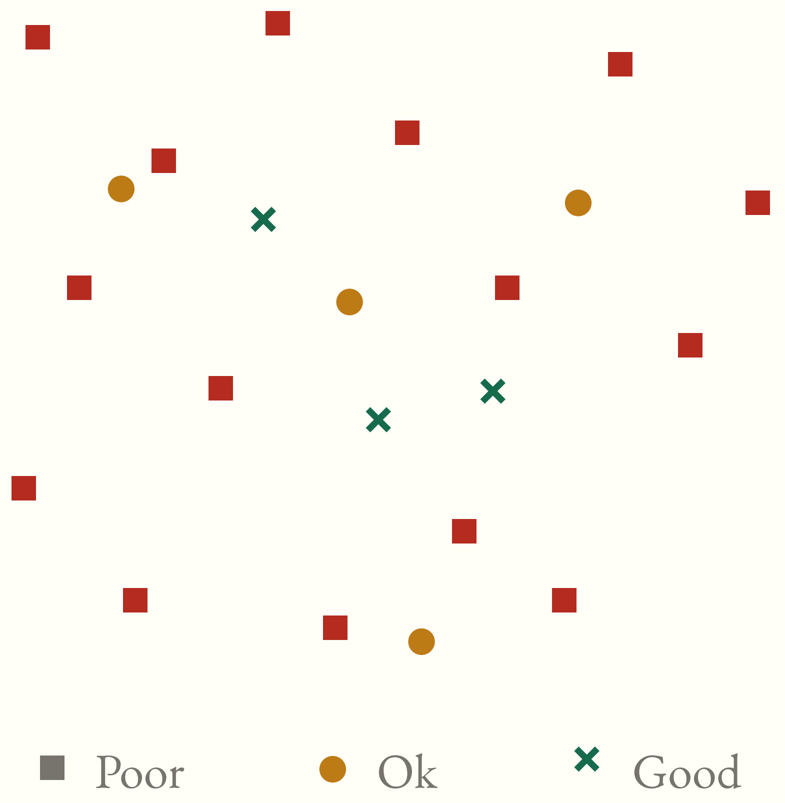
*Every task has countless possible designs, most of them are bad, and you can't tell which from a distance.*

# Space of Possible Solutions



# Space of Possible Solutions

# Space of Possible Solutions

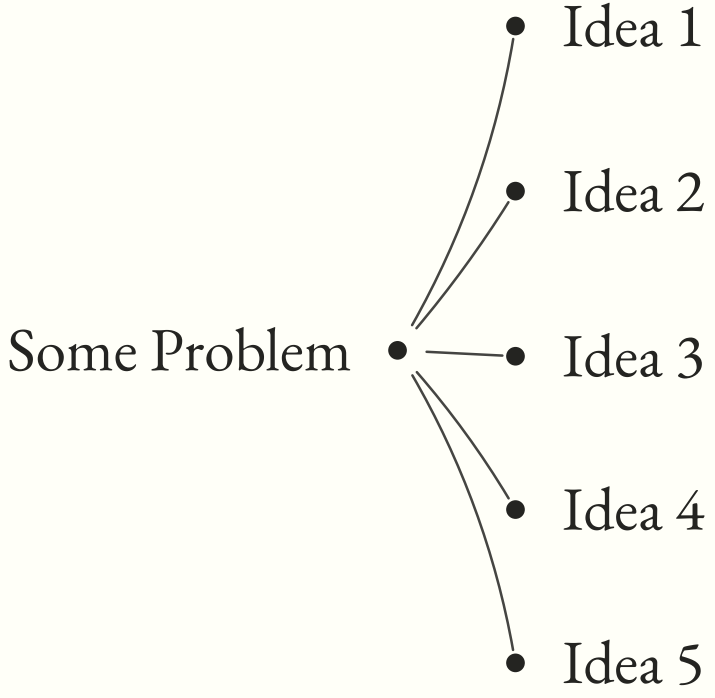


> *The problem of a small consideration space is the higher probability of settling for an OK solution and missing a good one. A fundamental principle of design is to* ***consider multiple alternatives*** *and then choose the best, rather than fixating on the first idea. One way to guarantee this:* ***explicitly generate multiple ideas*** *in parallel.*

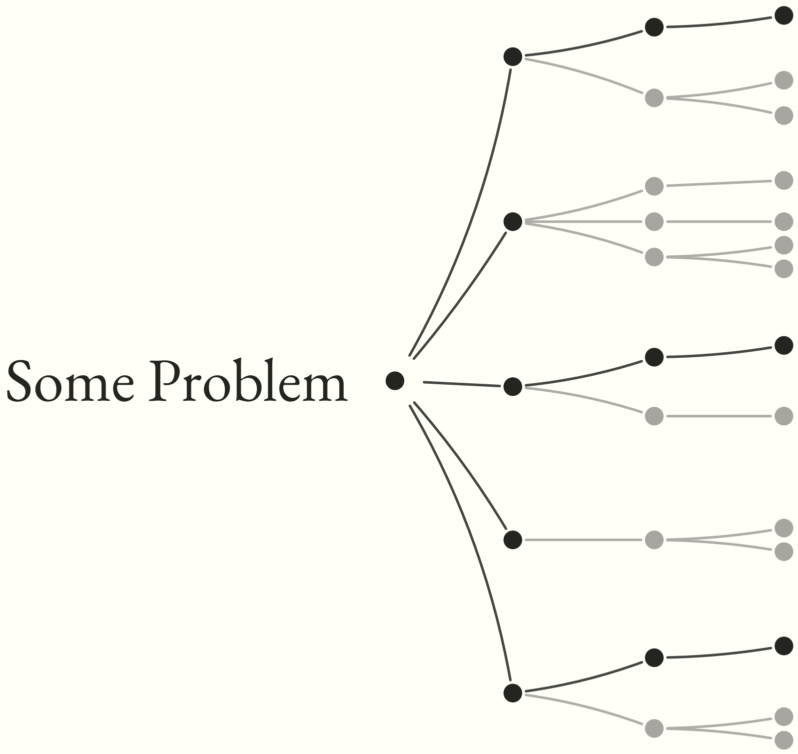
*— Tamara Munzner*

Visualization Analysis & Design, 2014

# Ideation as a Tree



# Ideation as a Tree



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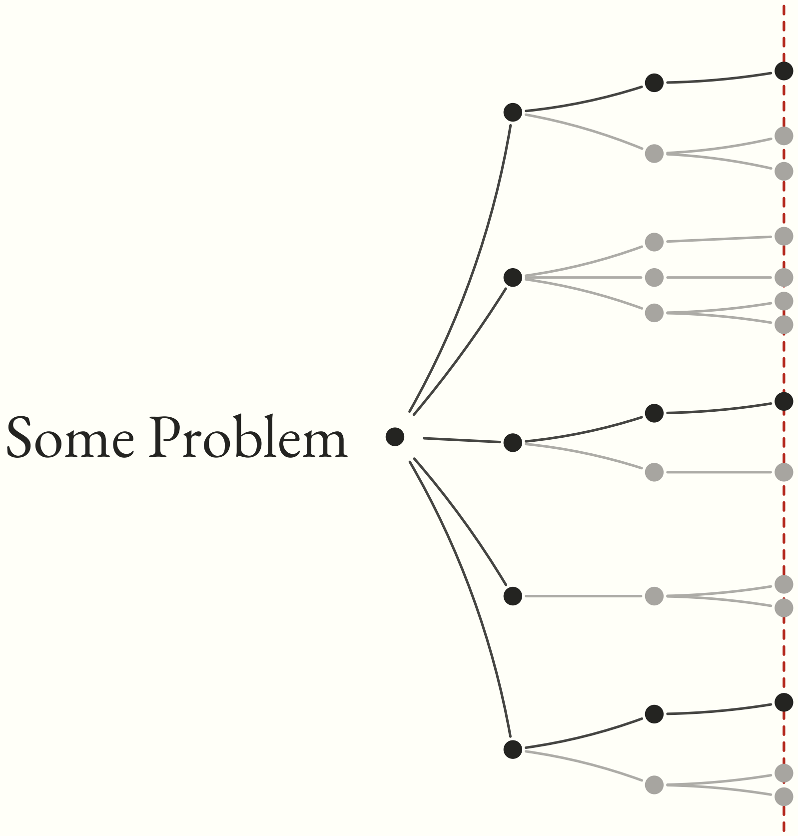

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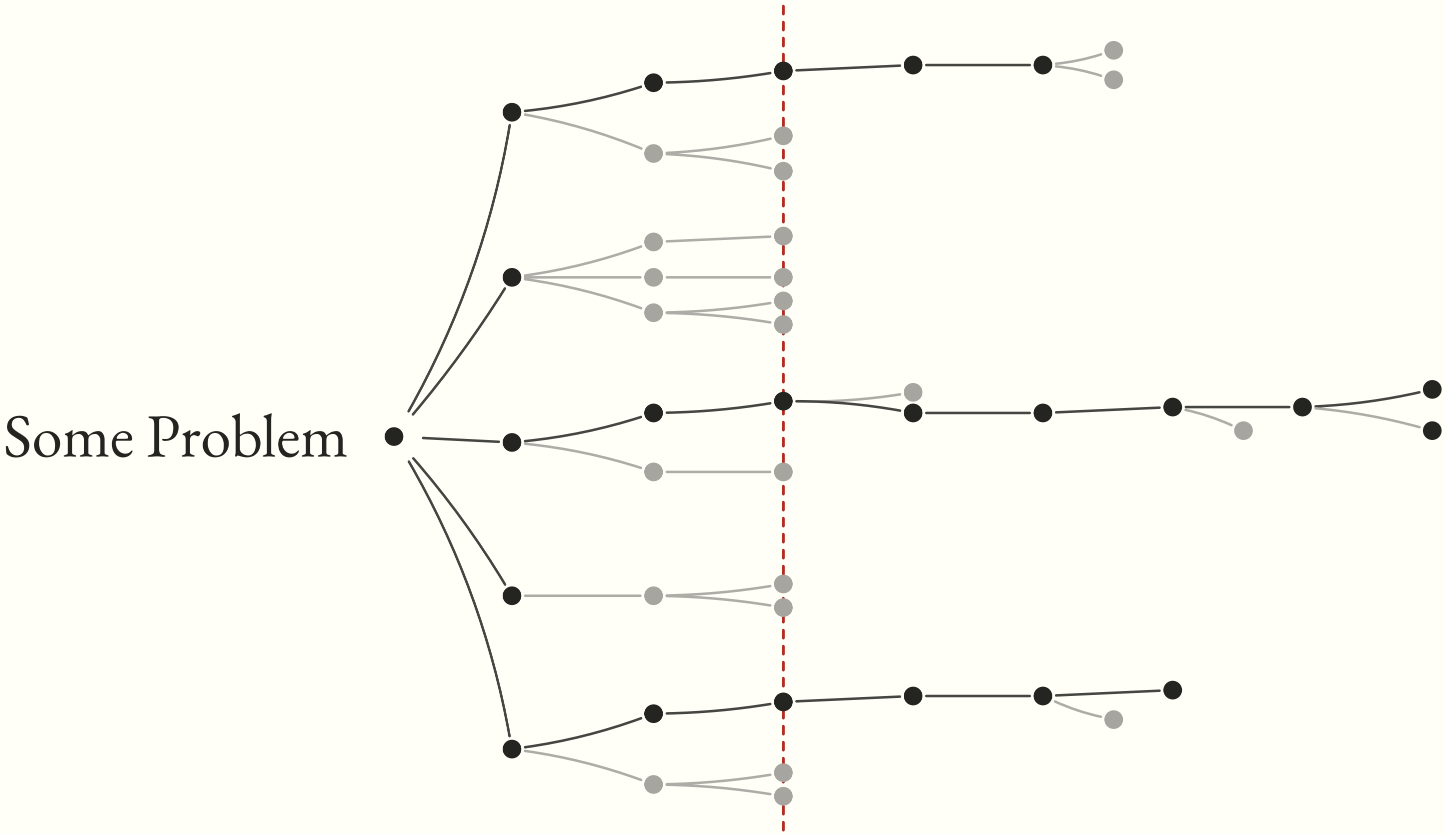
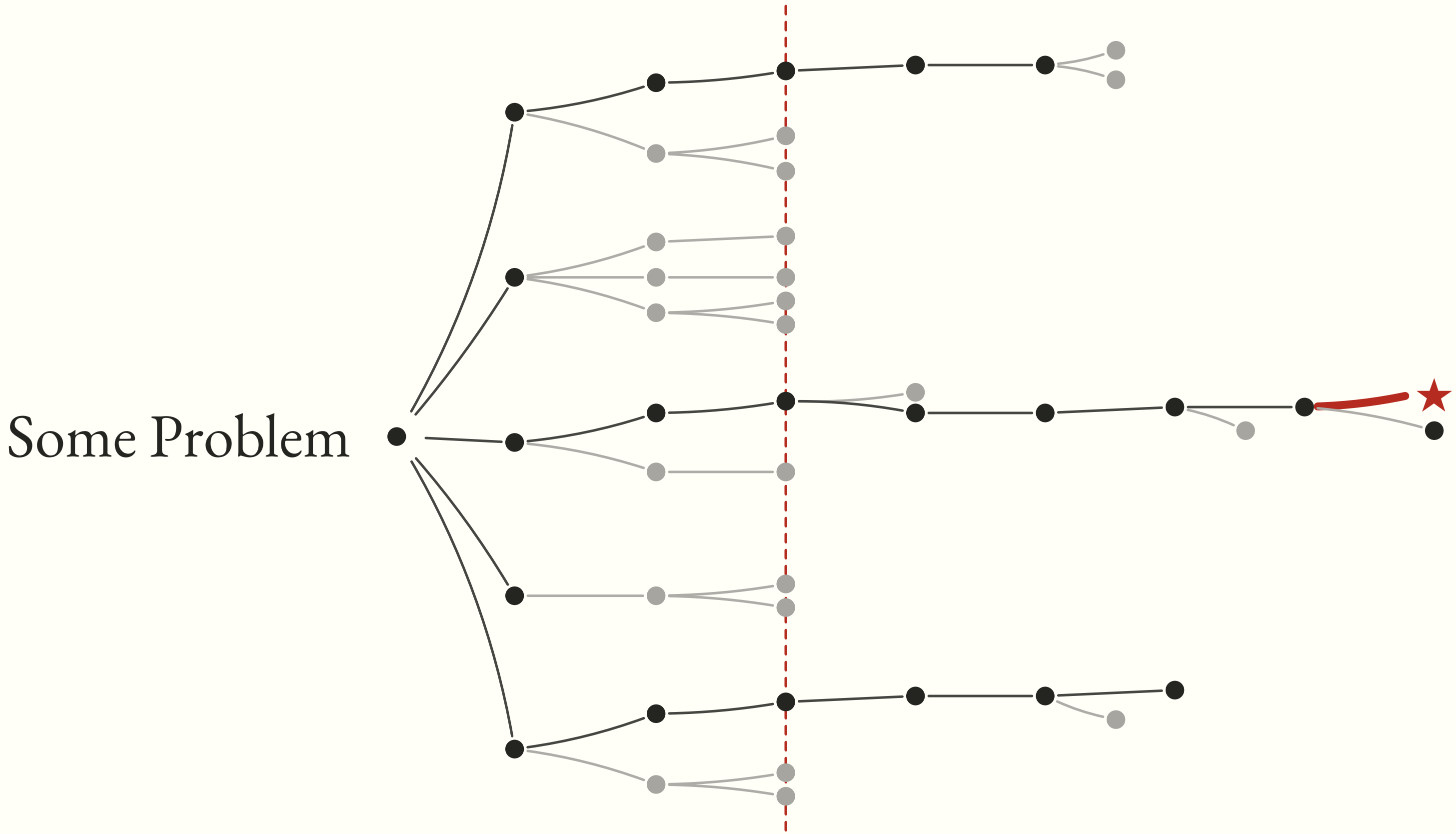


# Ideation as a Tree

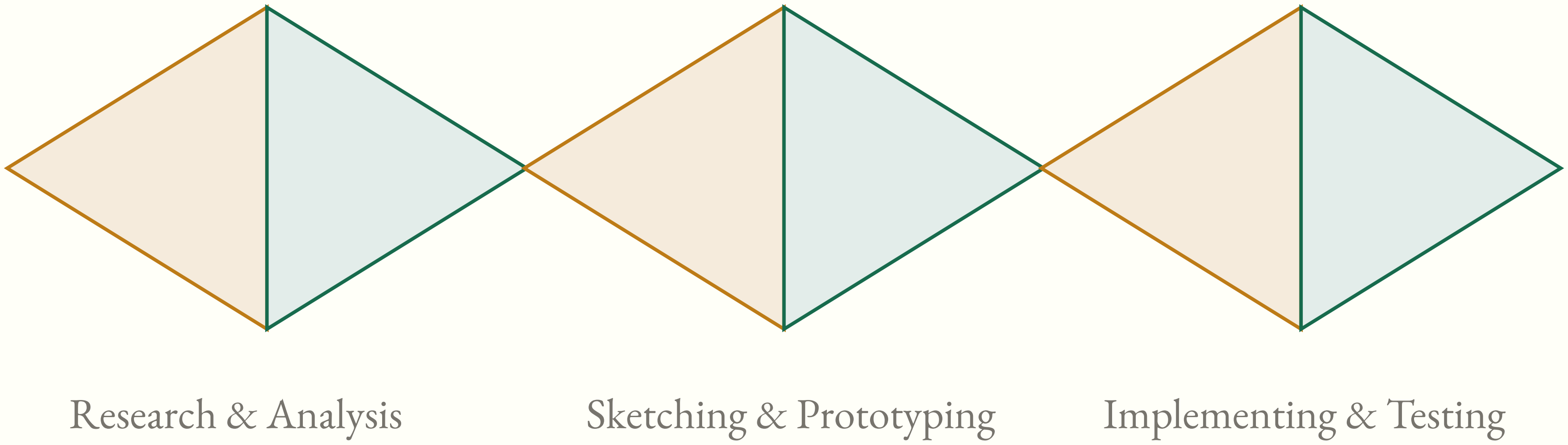


> *The best way to have a good idea is to **have lots of ideas**.*

*— Linus Pauling*

Professor of Chemistry, Caltech · UC San Diego · Stanford — only person awarded two unshared Nobel Prizes

# Elaborate and Reduce



> *Sketching is **fundamental to ideation and design**. Traditional disciplines — industrial design, graphic design, architecture — make extensive use of sketches to develop, explore, communicate, and evaluate ideas.*

— ***Tohidi, Buxton, Baecker & Sellen***

*User Sketches: A Quick, Inexpensive, and Effective Way to Elicit More Reflective User Feedback*, NordiCHI 2006

> *Sketching has long been a **best practice** for designers. Through sketches, designers follow a generative process of **developing, honing**, and **choosing ideas**. Designers also use sketches to discuss, exchange, and critique ideas with others.*
>
> — *Greenberg, Carpendale, Marquardt & Buxton*
>
> *Sketching User Experiences: The Workbook*, 2012

> *Sketches allow for a **dialog between the sketch and the viewer** — even when the viewer is the sketcher themself — that **facilitates better understanding** of the problem, and in turn, generation of new ideas.*

*— Tohidi, Buxton, Baecker & Sellen*

NordiCHI 2006

## *Sketching is **not about drawing!***

*Often the best sketches are just rough line drawings.*

# Sketching is a Conversation with the Self